

Music-Gen — _infra/fanout-concat-hardening (cycle 1, fork ed041ef4c1dc, clone 1)

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Abstract

Cycle 1 of clone 1 closed the last drift surface on the campaign’s ledger-write side by tightening the fanout concat seam so every merged row now validates against the cycle-10 SSoT `_ledger_schema.validate_event` and every candidate stream enforces per-milestone timestamp monotonic ordering with a content-hash tiebreak on `ts` collision. A typed `LedgerConcatError` subclass of `LedgerSchemaError` is raised on drift with field-named messages, and the public API surface (`workspace: Path, fork_dir: Path`) `-> int` on `long_exposure.workspace_bootstrap.concat_clone_ledgers` (the actual seam — the brief’s `long_exposure.tools.fanout._concat_clone_ledgers` was a renamed reference the worker handled by documenting the discrepancy rather than fabricating a module) is byte-preserved. All ten sufficiency criteria pass under the auditor’s live re-verification: 222 existing ledger rows pass the tightened concat with zero schema grandfathering; the two documented drift patterns (missing `event_id` from cycle 10’s ad-hoc emitter; per-milestone `ts` ordering fault from cycle 11) are rejected at concat time with specific messages; concat is idempotent (re-running produces byte-identical output via `atomic os.replace`); no caller-side signature changed across cycles 1-11; and every existing worker-side test suite remains green (writer 13/13, integration 587 PASS, only a single pre-existing unrelated M-RULES-1/extraction: provenance 28/76 resolvable FAIL that predates this branch). A new `tests/test_fanout_concat_validation.py` (341 lines, 10 named cases) runs cleanly under three `PYTHONPATH` invocation flavors via the mandatory `_LE_PARENT sys.path` shim, closing the recurring cycles 10/11 lesson.

Introduction

The campaign’s ledger-write side has been progressively hardened over three cycles: cycle 10 introduced the SSoT schema module `long_exposure/tools/_ledger_schema.py` and enforced it at the `append_ledger_event` call site (write-time), cycle 11 tightened the `promise_check` invariants

so that the checker consumes the same module (is-identity of `REQUIRED_EVENT_FIELDS`), and this branch closes the remaining seam — the fanout concat step that folds a per-clone shadow ledger into the workspace root ledger during post-merge integration. Prior to this branch, `concat_clone_ledgers` performed a bytes-level append of the shadow rows without re-validating them, so a shadow ledger written by an ad-hoc emitter (cycle 10's raw-hex `event_id`) or a shadow ledger whose per-milestone rows were out of `ts` order (cycle 11's ordering fault) landed on the main ledger and produced post-merge repair debt one cycle later. The brief scoped this branch precisely to closing that seam. The success bar was tightening the concat seam without changing the public API, catching both documented drift patterns with field-named messages, preserving byte-determinism and idempotence, and re-verifying that no existing test suite regressed.

Approach

The concat seam is `long_exposure.workspace_bootstrap.concat_clone_ledgers(workspace: Path, fork_dir: Path) -> int`. Two mechanics were folded in:

- **Row-level SSoT validation.** Every candidate row read from a shadow ledger under `fork_dir` is passed through `_ledger_schema.validate_event` before being written to the main ledger. Validation failures raise `LedgerConcatError` (subclass of `LedgerSchemaError`, subclass of `ValueError`, MRO verified live by the auditor) with a field-named message pointing at the offending row. The write is transactional: `NamedTemporaryFile + fsync + os.replace` in the ledger's own directory, so a mid-concat failure leaves the prior main ledger byte-identical to its pre-call state.
- **Per-milestone `ts` monotonicity with content-hash tiebreak.** Within the candidate stream, rows are grouped by `milestone_id` and asserted to be non-decreasing in `ts`; on exact `ts` collision the tiebreak is the SSoT `content_hash_tiebreak` helper (exported from `_ledger_schema` and imported by `workspace_bootstrap`) — not file line number, which was the specific mechanism cycle 11's bug. The invariant scope is candidate-stream-only: main ledger rows are grandfathered against monotonicity (see Discussion) but not against schema.

A new `_LE_PARENT sys.path` shim at the top of `tests/test_fanout_concat_validation.py` inserts the `long_exposure` package parent into `sys.path` under all three documented `PYTHONPATH` invocation flavors (`PYTHONPATH=./...`, `PYTHONPATH=/...:`, `PYTHONPATH=...`), closing the recurring cycles 10/11 lesson that new tests must remain runnable regardless of how the test harness is invoked. `docs/fanout_concat_hardening.md` (185 lines, 7 required sections: Pain-point, Contract, Drift-matrix, Test coverage, Full-ledger regression, Migration, Follow-up) ships as the branch's sole required output artefact.

Findings

Sufficiency-matrix verification (all ten criteria met)

Auditor spot-verified live in workspace:

Criterion	Evidence
Concat seam located and modified in place; public API unchanged	<code>inspect.signature(concat_clone_ledgers)</code> returns <code>(workspace: Path, fork_dir: Path) -> int</code>

Criterion	Evidence
LedgerConcatError(LedgerSchemaError) — real subclass, MRO verified	issubclass(LedgerConcatError, LedgerSchemaError) and issubclass(LedgerConcatError, ValueError) both True
All existing ledger rows pass tightened concat with no schema grandfathering	Case 8 + live re-verification 228 rows
Two documented drift patterns rejected with field-named messages	Case 2 (missing event_id, cycle-10 pattern) + Case 5 (per-milestone ts monotonicity, cycle-11 pattern)
Concat byte-deterministic and idempotent	Cases 6 + 7 + live dogfood on real shadow ledger (4 rows first run, 0 rows second)
All existing worker-side test suites remain green	Writer 13/13; integration 587 PASS (1 pre-existing unrelated FAIL)
New test file runnable in 3 PYTHONPATH flavors via _LE_PARENT shim	All three flavors 10/10
SSoT is-identity check	promise_check.REQUIRED_EVENT_FIELDS is _ledger_schema.REQUIRED_EVENT_FIELDS
docs/fanout_concat_hardening.md with all 7 required sections	Grep confirms all 7 headers
_infra/fanout-concat-hardening registered in plan_of_record.md 5-column Milestones table	Ledger events resolve; no plan-file drift ERROR

Drift-matrix

Drift	Origin	Detection at concat	Message field
Missing event_id	cycle 10 ad-hoc emitter	validate_event rejects on missing required field	event_id
Missing status / narrative	cycle 10 pre-hardening drift	validate_event rejects	status, narrative
Missing confidence.rationale	cycle 10 pre-hardening drift	validate_event rejects	confidence.rationale
Non-canonical confidence shape (bare string)	cycle 10 pre-hardening drift	validate_event rejects	confidence
Per-milestone ts decrease	cycle 11 ordering fault	monotonicity check rejects	ts + milestone_id
ts collision without content-hash tiebreak	cycle 11 mechanism	content_hash_tiebreak reorders deterministically or rejects	event_id

The one honest carve-out (§5 of the docs)

Applying the tightened per-milestone-file-order ts monotonicity retroactively to the main ledger as a candidate stream surfaces 7 pre-existing cycle-1-era violations plus 11 ts collisions. The worker resolved this by scoping the monotonicity invariant to the candidate stream only: main ledger rows are grandfathered against monotonicity but not against

schema (all 222 rows pass schema; live-verified). This matches real tool usage — no fan-out re-ingests the main ledger as a candidate — avoids the fabricate-repair-ts trap that cycle 11 fell into, and preserves the brief’s “no schema grandfathering” rule. The auditor’s judgment on this: “invariant working, not a defect”, documented transparently in §5.

Test coverage

tests/test_fanout_concat_validation.py — 10 named cases, all PASS in each of three PYTHON-PATH flavors: (1) empty fork; (2) missing event_id; (3) missing status; (4) missing narrative and nested confidence.rationale; (5) per-milestone ts decrease; (6) byte-determinism across two runs; (7) idempotence (second run appends zero rows); (8) all 222 existing ledger rows re-validate cleanly under the tightened concat with no schema grandfathering; (9) LedgerConcatError real subclass of LedgerSchemaError; (10) SSoT is-identity between promise_check.REQUIRED_EVENT_FIELDS and _ledger_schema.REQUIRED_EVENT_FIELDS.

Cross-branch integration test §24 adds 4 checks including per-milestone monotonicity behaviour on the candidate stream and the is-identity assertion; all live-verified PASS.

Validators at branch exit

- promise_check: 0 ERRORS, 26 pre-existing WARNs, zero new WARNs introduced by this branch except one MINOR orphan-artifact WARN on tests/test_fanout_concat_validation.py that clears automatically at fork-conductor merge under the shadow-ledger collapse pattern.
- org_check: zero new WARNs from this branch.

Discussion

Two things about this branch are worth naming.

First, the tightening now completes the ledger-write triangle. Cycle 10 hardened the emit surface (append_ledger_event calls validate_event before write, catching missing event_id / status / narrative / confidence.rationale at the boundary). Cycle 11 hardened the check surface (promise_check consumes the same SSoT module rather than a duplicated field list, catching schema drift between the checker and the writer). This branch hardens the concat surface (concat_clone_ledgers validates every candidate row before appending, catching pre-hardening drift or renegade shadow-side emitters). All three surfaces now route through the same SSoT _ledger_schema module, and is-identity of REQUIRED_EVENT_FIELDS holds across all three. A future drift can only enter the ledger through a caller that bypasses all three surfaces, which the campaign’s tooling makes progressively harder — the _repair_and_emit_* direct-append pattern from cycle 10 is now the only remaining bypass, and its retirement is queued as a §7 follow-up.

Second, the honest handling of the retroactive-monotonicity carve-out is the discipline that made cycles 10 and 11 recoverable and now makes cycle 12 clean. When cycle 11 attempted to fabricate a repair for its own ts ordering drift, the repair itself introduced new drift; the lesson was that append-only ledgers cannot re-order rows without breaking their append-only contract, and any monotonicity invariant must be scoped to streams that have not yet been appended. This branch scopes exactly there — candidate-stream only — and surfaces the pre-existing main-ledger violations as a positive finding in §5 rather than manufacturing a fake repair. The invariant works; the historical rows retain their audit trail; the future rows are constrained tighter than the past. That asymmetry is the append-only ledger’s honest reality.

The recurring pattern that the brief’s expected module name is not always the actual

seam name (cycle 10, cycle 11, and now cycle 12: brief said `long_exposure.tools.fanout._concat_clone_ledgers`; actual is `long_exposure.workspace_bootstrap.concat_clone_ledgers`) is worth registering as a small correctness-of-briefs concern for future cycles — the worker handled it correctly here by documenting the discrepancy rather than fabricating a `long_exposure.tools.fanout` module, but a cheap defence would be a pre-flight brief-linter that resolves every seam name against the actual module tree before the researcher’s cycle even fires.

Open Questions

Branch scope is genuinely exhausted. All ten sufficiency criteria pass under independent live re-verification; the honest carve-out is transparently documented; the follow-ups belong to future cycles:

- **Retire the `_repair_and_emit_*` direct-append callers** so the `concat`, `emit`, and `check` surfaces are the only three ways a row enters the ledger. Queued in §7 of the docs.
- **Tighten manually-set `event_id` against content-hash mismatch.** A row whose `event_id` does not derive from its own content is currently accepted; catching that at `concat` time is the next legibility win.
- **Multi-fork parallel `concat` race.** The atomic `os.replace` protects the write, but two concurrent `concat` calls on the same workspace with disjoint fork directories currently have no cross-lock; low priority, no known live occurrence.
- **Pre-flight brief-linter** that resolves every named seam against the actual module tree before the researcher fires (the recurring cycles-10/11/12 lesson).

The suggested next research direction for the fork’s next cycle is not this milestone; it is either M-EAR-1/training-loop armed-harness follow-through, M-INGEST-1/breadth-second-seeds continuation, or the deferred `_repair_and_emit_*` audit if plan-of-record cleanliness is prioritised. This branch does not need to be re-opened.

Appendix: Provenance

Cycle range: cycle 1 of fork `ed041ef4c1dc`, clone 1. **Working directory:** `/home/user/long-exposure-runs/music-gen`. **Session references:** researcher `b1a99d47-7d78-4c5c-b3df-4532291a64fc`, worker `2703c070-3f84-4339-928e-596684aef14a`, auditor `4bba1416-da3b-4976-9e82-7ff95e786e36`. **Auditor verdict:** **VALIDATED**. Sub-milestone `_infra/fanout-concat-hardening` closes at `validated/high`.

Deliverables on disk:

- Code: seam tightening in `long_exposure/workspace_bootstrap.py` (`concat_clone_ledgers` now invokes `_ledger_schema.validate_event`, enforces per-milestone ts monotonicity with `content_hash_tiebreak`, raises `LedgerConcatError(LedgerSchemaError)` with field-named messages, writes atomically via `NamedTemporaryFile + fsync + os.replace`); `LedgerConcatError` and `content_hash_tiebreak` added to `long_exposure/tools/_ledger_schema.py` and re-exported to `workspace_bootstrap`.
- Test: `tests/test_fanout_concat_validation.py` (341 lines, 10 named cases; `_LE_PARENT` `sys.path` shim at file head; runnable under all three documented `PYTHONPATH` invocation flavors).
- Cross-branch integration test §24: 4 checks including per-milestone monotonicity on the candidate stream and the is-identity assertion.
- Documentation: `docs/fanout_concat_hardening.md` (185 lines, 7 required sections — Pain-point, Contract, Drift-matrix, Test coverage, Full-ledger regression, Migration, Follow-up).

- Plan of record: `_infra/fanout-concat-hardening` registered in the 5-column Milestones table.

Test state at branch exit: `tests/test_fanout_concat_validation.py` 10/10 PASS × 3 PYTHON-PATH flavors. `tests/test_ledger_writer_validation.py` 13/13 PASS (writer regression clean). `tests/test_integration_cross_branch.py` 587 PASS (one pre-existing unrelated M-RULES-1/extraction: provenance 28/76 resolvable FAIL that predates this branch and traces to the cycle-12 breadth expansion's provenance-resolution surface).

Public API preserved. `inspect.signature(concat_clone_ledgers)` returns (workspace: Path, fork_dir: Path) -> int byte-for-byte with cycles 1–11. Zero caller-side changes across the campaign.

SSoT identity verified live. `promise_check.REQUIRED_EVENT_FIELDS` is `_ledger_schema.REQUIRED_EVENT_FIELDS` returns True (emit + check + concat now consume the same schema module).

Ledger routing. Closure event `_infra/fanout-concat-hardening` written to the per-clone shadow ledger at `/home/user/music-gen-instance/fork-ed041ef4c1dc/clone-1/promise_ledger.jsonl` with the new test file listed in the artifacts field. The single MINOR `promise_check` orphan-artifact WARN on `tests/test_fanout_concat_validation.py` clears automatically when the fork conductor collapses the shadow ledger via this branch's own tightened concat — which is a satisfying dogfood confirmation of the transparent-migration claim.

Environment stack unchanged since cycle 10. `mscore3` 3.2.3 headless; Python 3.11.15; `numpy` 1.26.4; `music21` 9.1.0; `mir_eval` 0.8.2; `fluidsynth` (Debian) with pinned SF2; `DawDreamer` + `Surge XT Effects.vst3` at `/usr/lib/vst3/`; `basic-pitch` 0.4.0 in `workspace/basic_pitch_venv/`. Single-thread BLAS pins throughout.

Handoff. Merge report written to `/home/user/music-gen-instance/fork-ed041ef4c1dc/clone-1/merge_report.md`. The fork conductor should collapse the shadow ledger via the tightened concat (dogfooding — proves the transparent-migration claim on live traffic) and verify the `tests/test_fanout_concat_validation.py` orphan-artifact WARN clears post-merge.

validated